

Smart warehouse technologies

- **Warehouse management system (WMS):** A well-implemented WMS consolidates critical data, offering real-time visibility and reporting. It improves demand forecasting, enhances labour efficiency, optimises warehouse layout, and boosts inventory accuracy.
- **Automated picking tools and guided vehicles (AGVs):** Modern warehouse operations increasingly rely on voice-guided systems and pick-to-light technology to ensure near-perfect picking rates and minimise errors, seamlessly integrating with WMS to enhance overall efficiency. Additionally, automated guided vehicles (AGVs) streamline storage and retrieval processes by navigating through digital paths for efficient loading and unloading. These self-guided vehicles offer scalability and easy integration with existing layouts, further enhancing warehouse operations.
- **Automated inventory control platforms:** Automation in inventory control eliminates manual logging and cycle counting, providing real-time, accurate stock reporting. This reduces errors, minimises disruptions, and ensures smooth warehouse operations.
- **IoT implementation:** By integrating sensors and data-driven insights, IoT technology optimises risk management, reduces theft, and improves both inventory control and labour planning. These systems integrate seamlessly with automated platforms, delivering comprehensive warehouse optimisation.

Embracing these technologies is vital to remaining competitive and thriving in today's logistics landscape.

printing across devices, enhancing accuracy and supporting key business functions such as sales tracking and bank transfers.

4. Wearable tech and exoskeletons. Warehouse operators, navigating fast-paced and decentralised environments under tight schedules, find efficiency in wearable technologies that allow hands-free mobility and access to comprehensive information. Smart glasses and intelligent gloves are particularly effective, enhancing productivity by ensuring tasks are performed with correct posture.

AGADE manufactures exoskeletons that adapt to various tasks, easing manual material handling by relieving muscular stress, especially on the spine, allowing warehouse employees to comfortably manage loads up to 25kg. Meanwhile, US startup Ox integrates machine learning and augmented reality into smart glasses, optimising inventory picking and replacing traditional handheld scanners. This enables real-time communication and automates workforce management and pick routing, significantly boosting productivity in

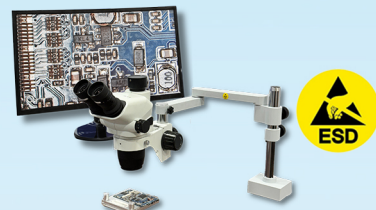
warehouse operations.

5. Enhancing worker safety.

Ensuring worker safety significantly enhances productivity and job satisfaction in industrial warehousing. Overlooking worker safety can result in reduced productivity, increased medical costs, and numerous workers' compensation claims. The implementation of technologies such as augmented reality (AR), machine vision, and wearables plays a vital role in promoting worker safety by continuously monitoring warehouse personnel. Furthermore, the integration of collaborative robots (cobots) and automated guided vehicles (AGVs) mitigates risks in hazardous environments, fostering a safer and more efficient workplace for employees.

Voxel, a US-based startup, has developed an AI-driven site visibility platform to enhance safety and eliminate near-miss incidents in warehouses. By leveraging existing CCTVs, the platform automates risk detection and reporting, providing ergonomic assessments to prevent musculoskeletal injuries. It also enables specific area control to de-

ESD Microscope JBM-EPL45



ESD: JBM EPL-45 Series microscope feature a specialized ESD Safe coating, protecting sensitive components from electrostatic discharges.

OPTICS: PL10X/Eyepieces, 22mm field of view

MAGNIFICATION: 0.67X – 45X and 186mm of working distance

LENSES: High-Quality achromatic lenses with anti-reflective coatings.

Digital 3D Microscope JBM-MZ3Di



Magnification : 0.7X-5x zoom(50x).

Working Distance : 100mm

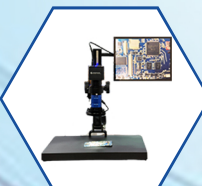
Sensor : Sony IMX307 (color)

In-Built Camera: 8 Mega Pixel



**Stereo Microscope
PCB Inspection
& Rework JBM-750T**

**Autofocus 4K Ultra HD
Microscope Camera
(8 MP) JBM-AF8**



**Digital
Microscope
JBM-MZ56**

JB INSTRUMENTS PVT.LTD.

ADD: J-9/36, S/F UNIT NO: 402-403, RAJOURI GARDEN, WEST DELHI, DELHI- 110027

MR. B.S BATRA : +91 79075 58994

MR. A.KRISHNA : +91 98735 71533

✉: info@jbinstruments.com